

Ikechukwu John Chukwu

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EDUCATION

Erasmus Mundus Joint Double MSc. in Cyber Security

Sept. 2023 – Sept. 2025

- Ss. Cyril and Methodius University, Skopje, North Macedonia
 - MSc. in Security, Cryptography and Coding
- Kadir Has University, Istanbul, Turkiye
 - MSc. in Cyber Security

B.Tech in Mathematics

Sept. 2013 - Sept. 2018

- Federal University of Technology, Nigeria

PUBLICATIONS

- *Algorithm for Key Transparency with Transparent Logs*. Open Research Europe, 4:163, 2024.
DOI: doi.org/10.12688/openreseurope.18200.2
- Detecting Obfuscated Malware on Windows Using Ensemble Learning Methods | Computer Science and Automation.
DOI: doi.org/10.15622/ia.24.1.5
 - Evaluated ensemble machine-learning methods for identifying obfuscated malware.
- Improving COVID-19 Detection Using Convolutional Neural Networks in Chest X-Ray Imaging.
DOI: doi.org/10.1117/12.3028812
 - Applied and evaluated convolutional neural networks for medical image classification.
- Integrating Cryptographic Techniques to Protect AI Systems and Data. | Proceedings of the 16th ICT Innovations Conference, pp. 54–70, 2024.
 - Led the analysis of cryptographic mechanisms for securing AI systems and data pipelines; evaluated homomorphic encryption, secure multiparty computation, and trusted execution environments.
- Image-Based Deep Learning Method For Effective Malware Detection
DOI 10.24412/2413-7383-106-122
- Securing AI Systems: A Comprehensive Overview of Cryptographic Techniques for Enhanced Confidentiality and Integrity. 13th Mediterranean Conference on Embedded Computing (MECO), pp. 1–8, 2024.
DOI: doi: 10.1109/MECO62516.2024.10577883.

RESEARCH EXPERIENCE

- **Cryptography Fellow @ MuseMatrix** Aug. – Dec. 2025
 - studied zero-knowledge proofs, secure multiparty computation, and homomorphic encryption, and gave a presentation on blockchain technology.
- **Masters Thesis @ Ss. Cyril and Methodius University, North Macedonia** Jan. – Aug. 2025
 - Topic: Assessing Lightweight Ciphers: An Evaluation Framework
- **Undergraduate Thesis @ Federal University of Technology, Owerri, Nigeria** 2018
 - Topic: Some Applications of Number Theory to Public-Key Cryptography

SKILLS & TRAINING

- **Languages:** Java, JavaScript, TypeScript, Solidity, Python, Go, SQL, Rust, Bash, HTML, CSS
- **Frameworks & Libraries:** React.js, Node.js, Next.js, Express.js
- **Cryptography:** Finite Fields, Provable Security, Formal Security Models, zk-SNARKs, zkPs, OpenFHE
- **Further Training:** Mathematics of Machine Learning (Coursera/Deeplearning.AI) (Certificate); University Teaching Certificate HKU/Coursera (Certificate); Applied Data Science Lab - World Quant University (Certificate)

AWARDS

- Erasmus Mundus Joint Master's Scholarship in Applied Cybersecurity (CyberMACS) 2023 - 2025
- ORSHIN Summer School on Open-source ReSilient Hardware and software for Internet of thiNgs, Greece Sept. 2025

TEACHING EXPERIENCE

- **Technical Instructor @ HIIT, Lagos, Nigeria - Link** Aug. 2022 – 2023
 - Taught HTML5/CSS3, Responsive design with a focus on Human Centered Interaction (HCI), JavaScript, Version Control with git/GitHub to over 300 students.
 - Supported over 300 students in debugging projects and improving problem-solving skills.
- **Mathematics Teacher @ Holy Rosary College, Enugu, Nigeria - Link** Aug. 2021 – 2022
 - Prepared over 250 students for the WAEC Mathematics examination, covering topics in Pre-calculus, Algebra 1, Trigonometry, and Probability Theory.
 - Skills gained: Classroom management, tutoring, grading exercises, and providing comprehensive feedback.
- Delivered a lecture on the permutation of Ascon @ Faculty of Computer Science, UKIM - Lecture Link.
- Delivered a lecture on Two Factor Authentication, Cookies, and SSO @ Faculty of Computer Science, UKIM - Lecture Link

PROFESSIONAL EXPERIENCE

- **Cyber Security Engineer @ Gigatt, Skopje, North Macedonia** Dec. 2024 - Present
 - Delivered cybersecurity awareness training and security updates to 3 staff.
 - Developed practical expertise in web application security and secure digital platform in an e-commerce store and three web projects.
- **Cyber Security Consultant @ KPMG, Istanbul, Turkiye** Jul. – Oct. 2024
 - Served as a consultant for a bank in Amman, Jordan, by assessing IT risks and analyzing data using the NIST framework. Supervisor: Betül Uzun
- **Quality Assurance Evaluation Expert @ HCERES, France** Sept. – Nov. 2025
 - Served as an external evaluator for the Cyberus joint program under the European Approach for Quality Assurance of Joint Programs.
 - Contributed two sections: Student Support Services and Transparency of Documentation, to the published evaluation report.
- **Quality Assurance Student Representative @ CyberMACS.eu, Turkiye** Sept. 2023 - 2025
 - Developed practical expertise in quality assurance methodologies, focusing on cybersecurity education across consortium institutions of the CyberMACS EU project. Certificate of Recognition

CERTIFICATIONS

- CompTIA Security+ ce (Certificate). June. 2026
- AWS Cloud Practioner (Certificate). Mar. 2026

PROJECT HIGHLIGHTS

- **Private KV Cache for Autoregressive LLM Inference under FHE** github
 - Implemented privacy-preserving transformer inference using the CKKS scheme and OpenFHE.
- **Lox Programming Language Interpreter & Virtual Machine | Java, C, Git** github
 - Designed and implemented a complete scripting language runtime from scratch, following the architecture described in Crafting Interpreters.